Experiment–1:

Operations on Classical Sets

Aim

To implement basic operations on classical (crisp) sets such as union, intersection, difference, and complement.

Theory

In classical set theory, an element either belongs to a set or does not belong to it. The basic set operations are:

- **Union**: Combines elements of two sets
- **Intersection**: Finds common elements
- **Difference**: Elements present in one set but not in the other
- **Complement**: Elements not present in the set with respect to a universal set

Algorithm

1. Define the universal set.
2. Define two classical sets.
3. Perform union and intersection operations.
4. Perform difference operations.
5. Find complements with respect to the universal set.
6. Display the results.

Program

```
# Define universal set
U = {1, 2, 3, 4, 5, 6}

# Define two classical sets
A = {1, 2, 3}
B = {3, 4, 5}

# Union
union_set = A.union(B)

# Intersection
intersection_set = A.intersection(B)

# Difference
difference_A_B = A.difference(B)
difference_B_A = B.difference(A)

# Complement
complement_A = U.difference(A)
complement_B = U.difference(B)

# Display results
print("Universal Set:", U)
print("Set A:", A)
print("Set B:", B)

print("\nUnion :", union_set)
print("Intersection :", intersection_set)

print("Difference :", difference_A_B)
print("Difference :", difference_B_A)

print("Complement of A:", complement_A)
print("Complement of B :", complement_B)
```

Output

```
Universal Set: {1, 2, 3, 4, 5, 6}
Set A: {1, 2, 3}
Set B: {3, 4, 5}

Union : {1, 2, 3, 4, 5}
Intersection : {3}
Difference : {1, 2}
Difference : {4, 5}
Complement of A: {4, 5, 6}
Complement of B : {1, 2, 6}
```

Conclusion

The basic operations on classical sets were successfully implemented.